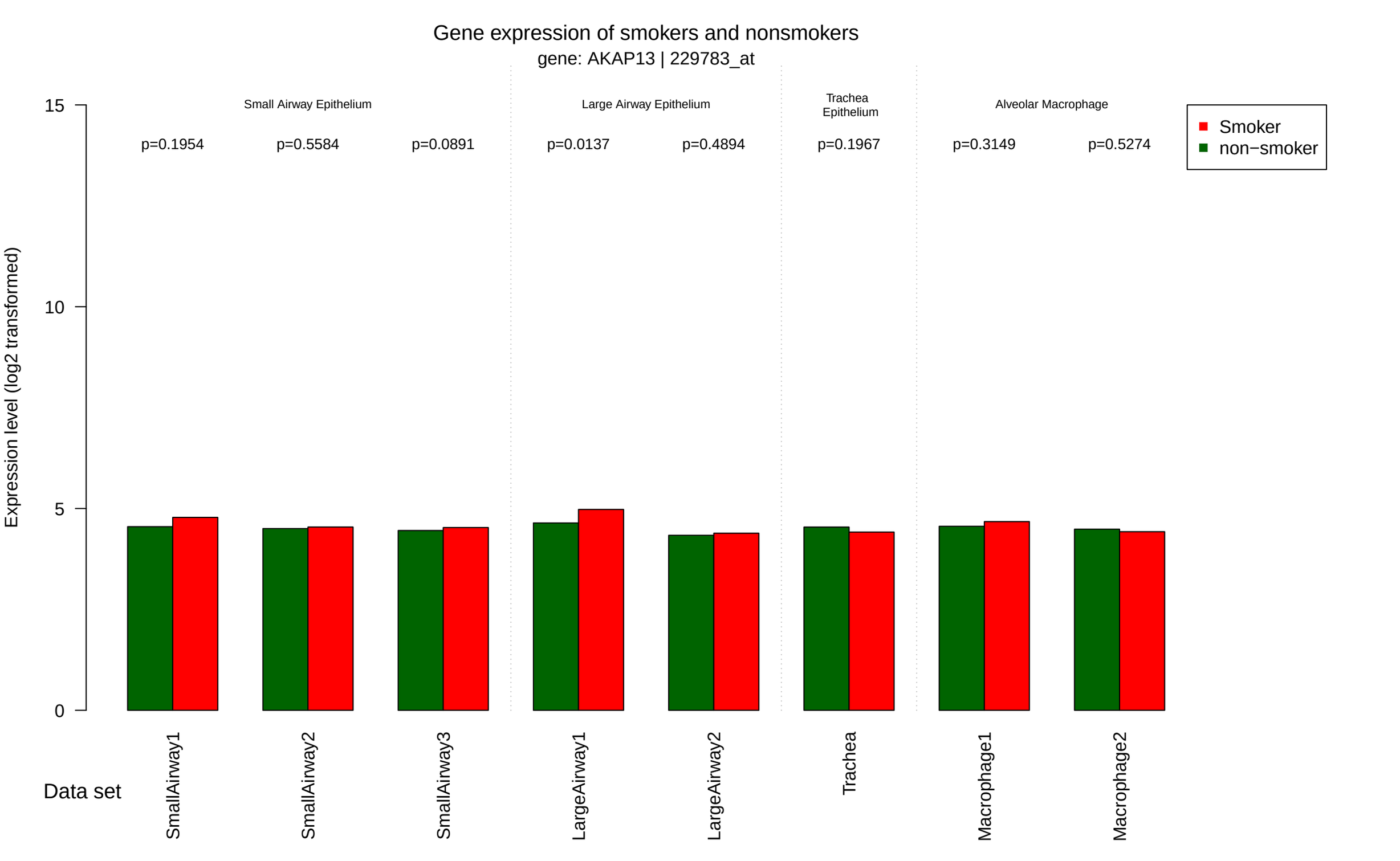
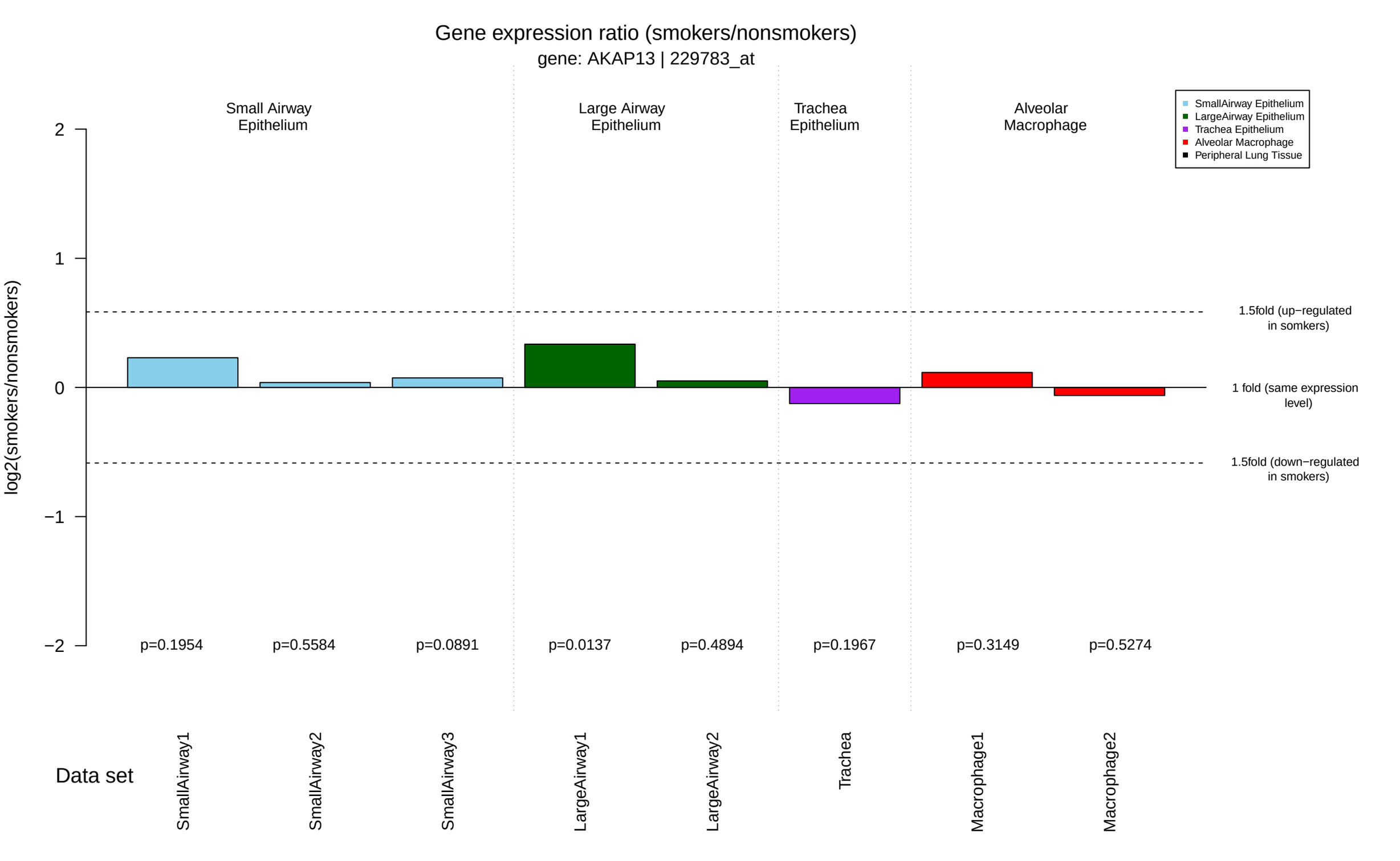
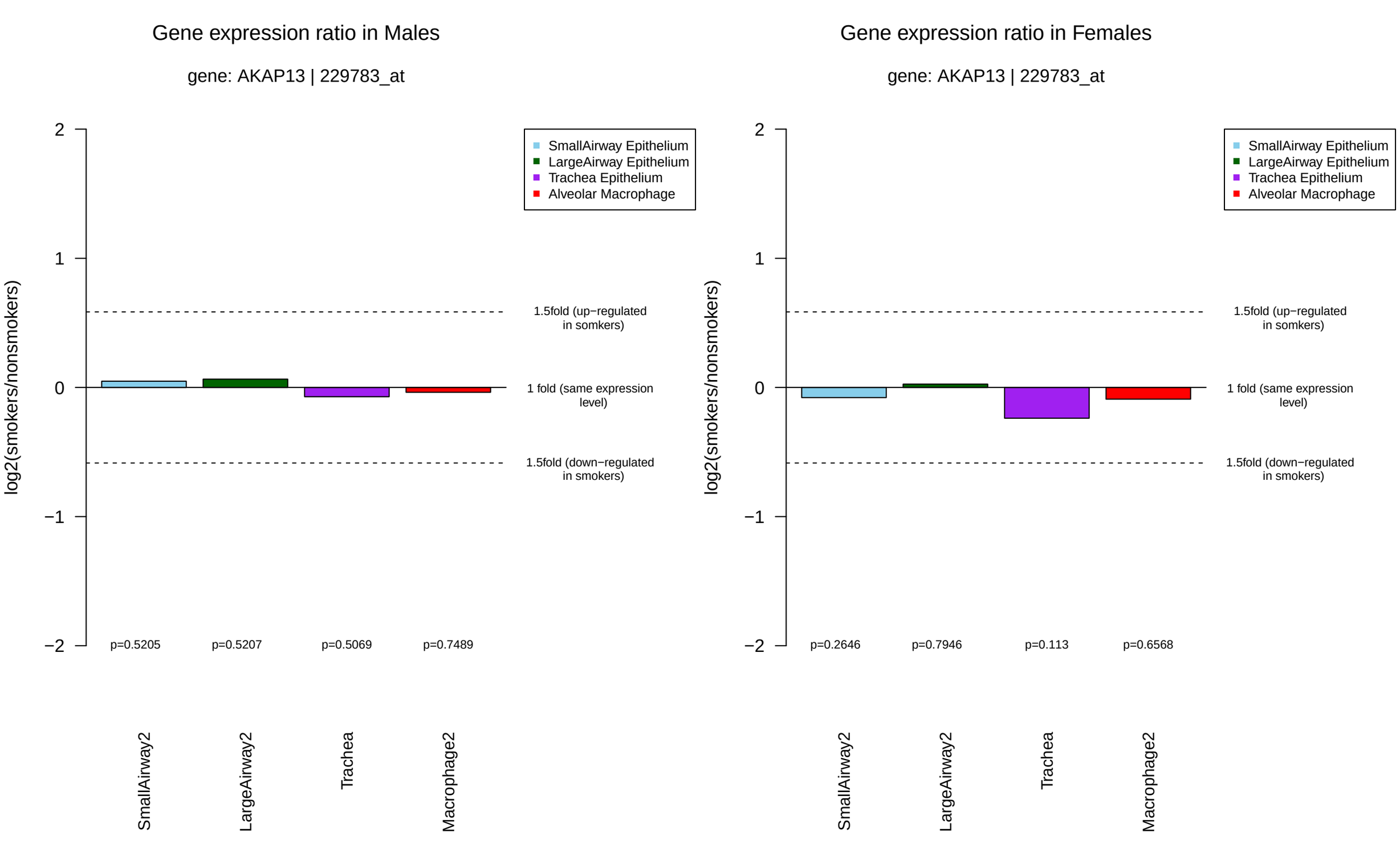
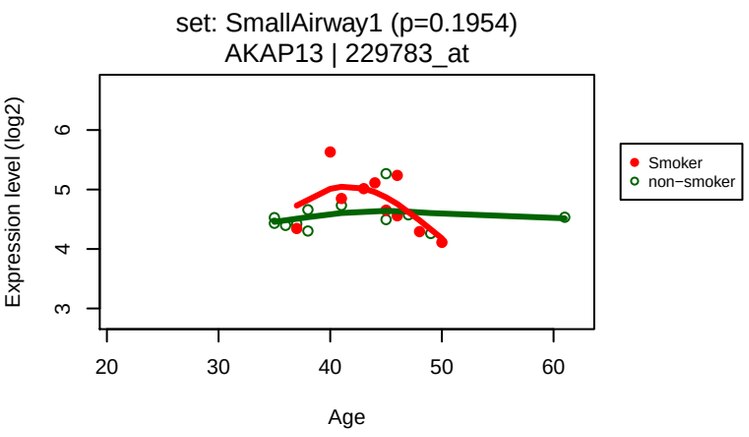
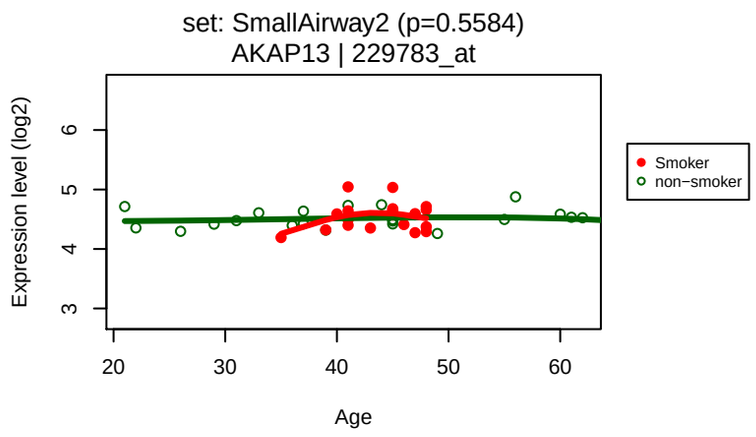
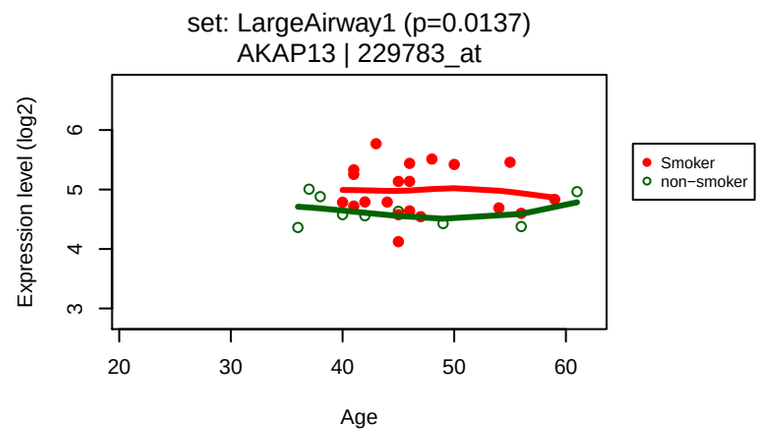

Gene expression of smokers and nonsmokers
gene: AKAP13 | 229783_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1954
p=0.5584
p=0.0891
p=0.0137
p=0.4894
p=0.1967
p=0.3149
p=0.5274
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

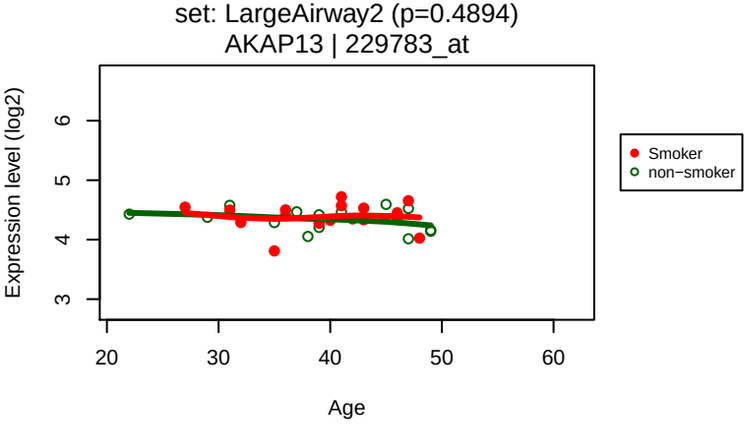

Gene expression ratio (smokers/nonsmokers)
gene: AKAP13 | 229783_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1954
p=0.5584
p=0.0891
p=0.0137
p=0.4894
p=0.1967
p=0.3149
p=0.5274
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

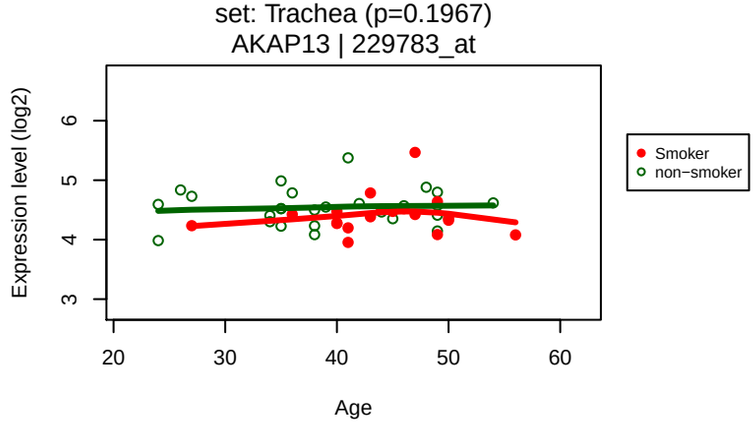

Gene expression ratio in Males
gene: AKAP13 | 229783_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5205
p=0.5207
p=0.5069
p=0.7489
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKAP13 | 229783_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2646
p=0.7946
p=0.113
p=0.6568
SmallAirway2
LargeAirway2
Trachea
Macrophage2

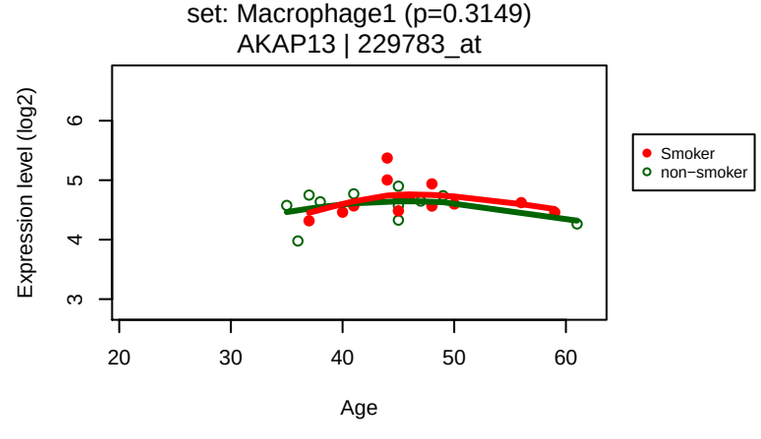

set: SmallAirway1 (p=0.1954)
AKAP13 | 229783_at
Expression level (log2)
Age
Smoker
non−smoker

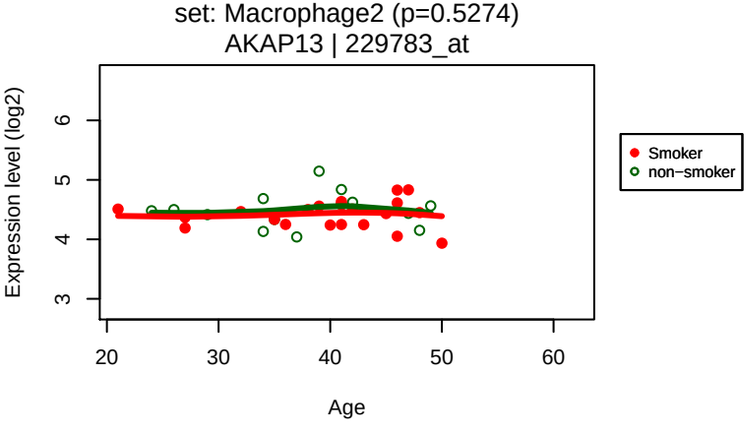

set: SmallAirway2 (p=0.5584)
AKAP13 | 229783_at
Expression level (log2)
Age
Smoker
non−smoker

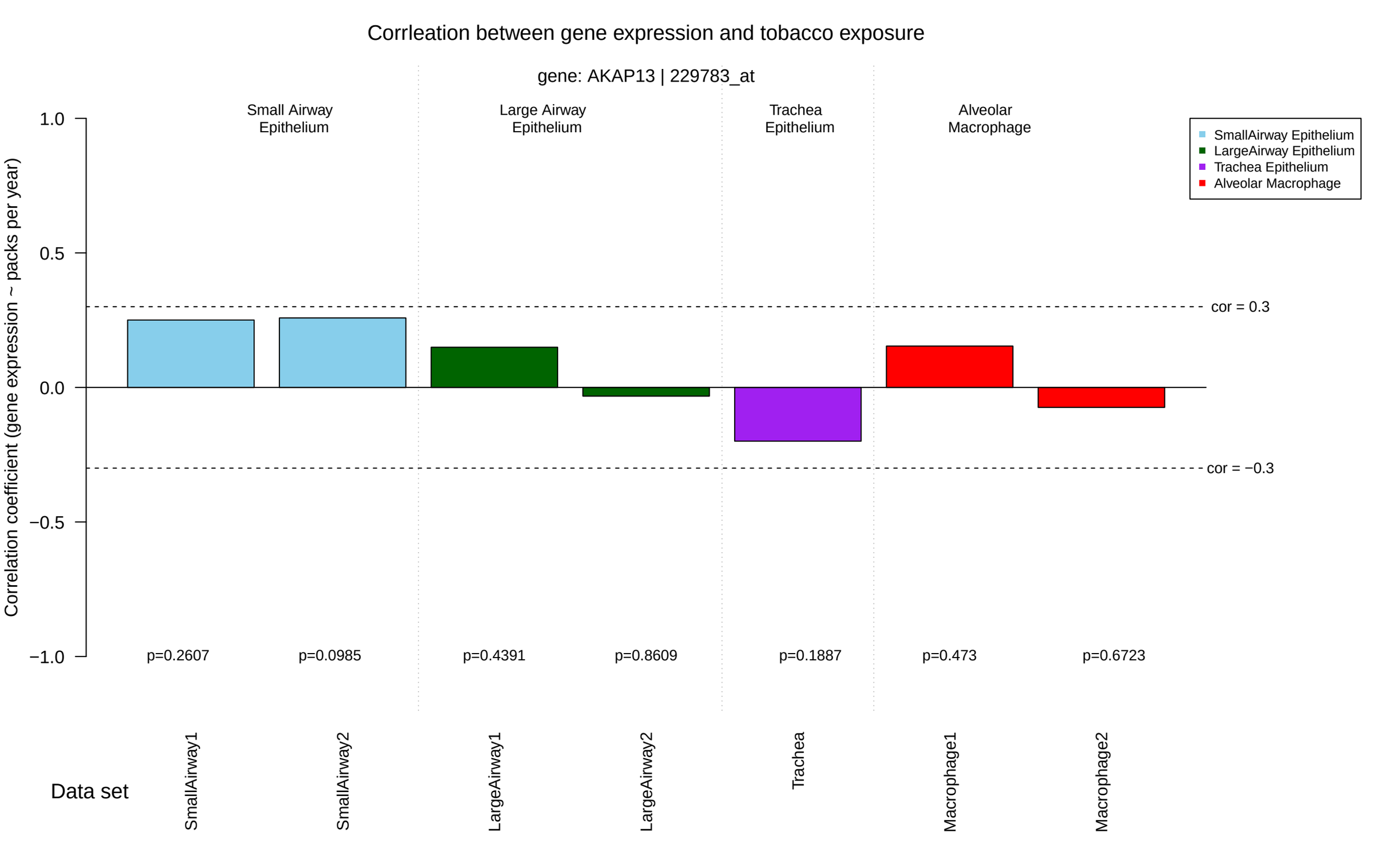

set: LargeAirway1 (p=0.0137)
AKAP13 | 229783_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4894)
AKAP13 | 229783_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1967)
AKAP13 | 229783_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3149)
AKAP13 | 229783_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5274)
AKAP13 | 229783_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKAP13 | 229783_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2607
p=0.0985
p=0.4391
p=0.8609
p=0.1887
p=0.473
p=0.6723
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2